

Article

Ranking Inversion in Risk-Parameterised UAV Path Planning for Wildfire Emergency Response

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Highlights:

- Planner orderings by navigation success rate diverge from orderings by mission effectiveness—a ranking inversion confirmed across 450 episodes (Friedman $p < 0.001$).
- A single risk coefficient ρ governs the survival–timeliness trade-off: higher ρ increases survival probability at the cost of time-induced mission degradation.
- Navigation success rate is an insufficient evaluation criterion for time-critical disaster-response missions in which delivery time determines objective value.
- The risk coefficient serves as a principled mission-selection parameter, enabling deployment-specific planner configuration.

Abstract: Existing path-planning benchmarks evaluate UAV planners by navigation success rate—a binary metric that disregards delivery time. This paper demonstrates that success rate is an insufficient and, in time-critical disaster-response settings, a misleading evaluation criterion: the planner with the highest survival probability can yield the lowest mission effectiveness. This phenomenon is termed *ranking inversion*. To characterise this effect, we introduce FLARE, a deterministic benchmark that evaluates five risk-parameterised planners across 450 episodes on three OpenStreetMap-derived Greek wildfire scenarios with coupled hazard dynamics (wind-driven fire, structural collapse, traffic closures). Each planner differs by a single risk coefficient ρ that controls sensitivity to a shared hazard cost map. FLARE quantifies mission impact—medication efficacy, casualty survival, surveillance freshness—using strictly decreasing functions of delivery time. Experimental results show that the planner with the highest success rate (62.7%) ranks last among adaptive planners in mission score ($\Delta\text{Rank} = -3$), whereas a risk-tolerant planner achieves the highest mission score despite ranking third in navigation ($\Delta\text{Rank} = +2$; Friedman $p < 0.001$). The underlying mechanism is identified: higher ρ induces longer detours that increase survival probability at the cost of time-induced mission degradation.

Keywords: UAV path planning; wildfire emergency response; ranking inversion; risk-aware planning; mission-impact evaluation; deterministic benchmarking; reproducibility

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1. Introduction

Natural disasters such as wildfires, earthquakes, and floods generate rapidly evolving hazard landscapes that severely disrupt ground-based logistics. In such events, the timely delivery of medical supplies, situational awareness, and casualty location constitutes the primary operational objective for emergency responders [1,2]. Traditional relief operations rely on manned helicopters and ground convoys; however, both modalities suffer from road network damage, airspace congestion, and limited scalability [3]. Unmanned aerial vehicles (UAVs) have emerged as a promising alternative, offering low-cost, rapid-deployment platforms capable of operating in degraded environments where manned access is infeasible [4,5].

Over the past decade, extensive research has addressed the UAV path planning problem across a range of algorithmic families: graph-search methods such as A* [6] and D* Lite [7], potential-field approaches [8], sampling-based planners, meta-heuristic optimisers, and, more recently, deep reinforcement learning agents [9–11]. Comprehensive surveys confirm that path planning remains one of the most actively studied problems in UAV autonomy, with over 3 500 publications indexed between 2000 and 2024 [12,13]. Risk-aware extensions that modulate edge costs by proximity to hazards have further improved planner robustness in cluttered environments [14–16].

Despite these advances, most existing evaluations share a common limitation: they assess planner performance using navigation-centric metrics—path length, success rate, or computation time—without accounting for the temporal value of the mission objective [17]. In disaster response, however, mission effectiveness degrades monotonically with elapsed time: insulin delivered after 200 steps carries substantially greater clinical value than the same delivery at 800 steps; a fire-perimeter survey completed before containment-line decisions is actionable, while one arriving afterwards is not. Navigation success is therefore a necessary but insufficient proxy for mission effectiveness.

This distinction is particularly acute in wildfire scenarios, where the hazard landscape expands continuously. Cellular-automaton (CA) fire models with wind-directional spread [18–20] generate time-coupled blocking dynamics that fundamentally alter the planning problem: corridors that are traversable at step t may be impassable at step $t + \Delta t$. Under such conditions, UAV planners face an intrinsic trade-off between survival probability and delivery speed. A conservative (high-risk-aversion) planner increases its likelihood of reaching the goal by circumnavigating hazard zones, but the resulting temporal penalty may render the completed mission ineffective. A risk-tolerant planner may attempt shorter routes through fire fringes, accepting higher collision probability in exchange for faster arrival. Recent multi-UAV firefighting studies confirm the complexity of planning under expanding fire dynamics [21,22], yet none evaluate the mission-level consequences of the survival–speed trade-off.

This paper identifies and characterises the *ranking inversion* phenomenon: planner orderings by navigation success rate diverge systematically from orderings by mission impact score. Across 450 deterministic episodes on three Greek wildfire scenarios, the planner with the highest survival rate ranks last in mission effectiveness.

The underlying mechanism traces to a single parameter—the risk coefficient $\rho \in \mathbb{R}_{\geq 0}$ —that governs how strongly each planner avoids hazards. Higher ρ amplifies perceived danger, producing wider detours with higher survival probability. Lower ρ discounts risk, permitting shorter-time routes through fire fringes at the cost of increased collision probability. Risk is not penalised directly—it is penalised through time-induced mission degradation:

$$\rho \uparrow \Rightarrow \text{higher } P(\text{survive}) \Rightarrow t \uparrow \Rightarrow M(t) \downarrow \quad (1)$$

where $t \in \mathbb{N}$ is the delivery step and $M : \mathbb{N} \rightarrow [0, 1]$ is the strictly decreasing mission score.

Contributions.

- **FLARE benchmark:** three OSM-derived wildfire scenarios with coupled hazard dynamics and time-decaying mission scores, grounded in real Greek incidents (2021 Evia, 2023 Evros), enforcing 36 deterministic contracts.
- **Risk-parameterised planner suite:** five planners spanning static, adaptive, and reactive families, differing solely in ρ .
- **Ranking inversion:** empirical demonstration, supported by statistical validation (Friedman $p < 0.001$, Shapley attribution), that planner orderings by navigation success rate diverge from orderings by mission effectiveness.

The remainder is organised as follows: Section 2 describes the benchmark design; Section 3 presents experiments; Section 4 interprets findings; Section 5 concludes.

1.1. Related Work

UAV path planning in disaster response. The deployment of UAVs for disaster relief has expanded rapidly, with applications spanning medical delivery, search-and-rescue, damage assessment, and communications relay [1,2,4]. Path planning constitutes a central enabling technology for these missions. Classical graph-search planners—A* [6], D* Lite [7]—and reactive controllers such as artificial potential fields (APF) [8] remain widely adopted due to their determinism and computational tractability. Recent surveys document a growing trend toward hybrid architectures that combine classical planning with learning-based adaptation [5,9–11]. Risk-aware extensions weight traversal costs by hazard proximity or probabilistic threat maps [14–17], demonstrating improved robustness in uncertain environments. However, these evaluations employ path-level metrics exclusively: path length, computation time, or binary success. No existing work evaluates the *mission-level* consequences of risk sensitivity—an omission that the present work addresses.

Reproducible benchmarking. Henderson et al. [23] demonstrated the fragility of reinforcement-learning evaluations when seeds and hyperparameters are not controlled. Subsequent best practices now include interquartile mean (IQM) with bootstrap confidence intervals [24], non-parametric Friedman/Wilcoxon rank tests [25], and performance profiles [26]. Bonsignorio et al. [27] advocate published experimental contracts as a reproducibility mechanism. A recent bibliometric analysis of the UAV path planning literature confirms that standardised evaluation protocols remain rare, with fewer than 8% of studies reporting seed-controlled deterministic experiments [12]. FLARE adopts all four elements: deterministic replay, IQM, rank tests, and 36 enforceable contracts verified by 189 automated assertions.

Dynamic fire environments. Cellular-automaton (CA) models with wind-directional spread [18–20] capture the dominant fire propagation mechanism and have been validated against satellite-observed burn patterns at landscape scale [28]. Multi-UAV planning under spreading fire has recently attracted attention: Wang et al. [21] address multi-target firefighting task allocation in dynamic forest fire environments, while Liu et al. [22] propose cooperative firefighting strategies under resource constraints using fire-spread-aware scheduling. Most UAV path planning benchmarks, however, employ static or randomly toggled blockages [29]—environments that do not worsen over time. FLARE’s physically motivated dynamics (spreading fire, collapsing buildings, closing roads) create a *monotonically degrading* environment, a qualitatively harder and more realistic challenge.

The missing evaluation paradigm. Across all three research threads—path planning, reproducible benchmarking, and dynamic environments—existing work shares one critical blind spot: *time-coupled mission value*. Existing evaluation frameworks count completions but do not assess their temporal value. By coupling a strictly decreasing mission score $M(t)$ with expanding hazard dynamics, FLARE reveals performance trade-offs that success rate alone cannot capture.

2. Materials and Methods

2.1. Environment

FLARE operates on a grid $\mathcal{G} = \{0, \dots, 499\}^2$ (~ 3 m per cell), extracted from OpenStreetMap tiles [30] of Greek urban areas. The Gymnasium-compatible [31] environment exposes $|\mathcal{A}|=5$ actions (four cardinal moves and stay). A strict timing contract governs each step: the planner observes the *current* blocking mask (frozen), selects an action, and only *then* do dynamics advance (fairness contract FC-2). If fire or debris reaches the agent’s cell after the dynamics update, the episode terminates immediately.

2.2. Hazard Dynamics

Three coupled layers produce a hazard landscape that worsens monotonically.

Wind-driven fire. Spread follows a cellular automaton on the Moore neighbourhood [18], modulated by wind direction and land use. Downwind cells ignite faster; up-

wind cells are suppressed (Figure 1). A buffer zone (radius 1) and dense smoke further restrict UAV movement.

Wind turns symmetric danger into a deadly corridor

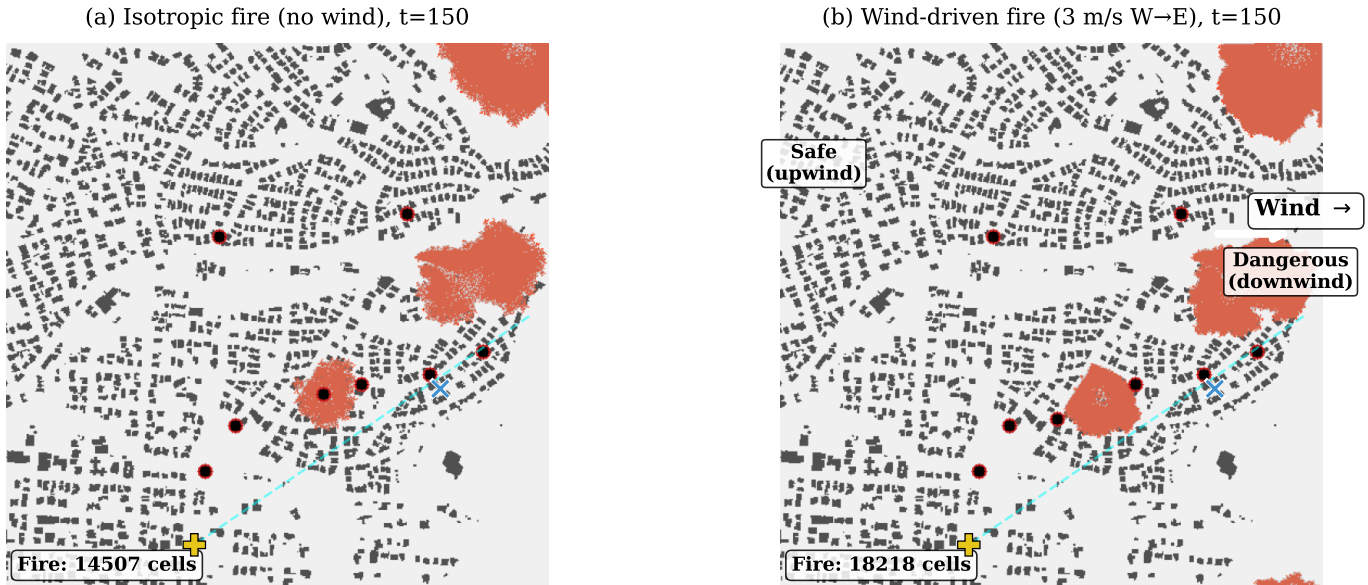


Figure 1. Wind reshapes the hazard geometry. (a) Without wind, fire expands symmetrically. (b) At 3 m/s from the west, fire elongates downwind, blocking the direct route while the upwind flank remains passable for risk-tolerant planners. Black circles denote dynamic obstacles; blue cross denotes the UAV. Penteli, seed 42, $t=150$.

Structural collapse. Buildings exposed to fire for $\tau_c=80$ steps collapse with probability $p_d=0.6$, scattering permanent debris. Unlike fire—which burns out—debris *accumulates*, progressively eliminating detour options (Figure 2).

Fire → 80 steps → collapse → permanent debris

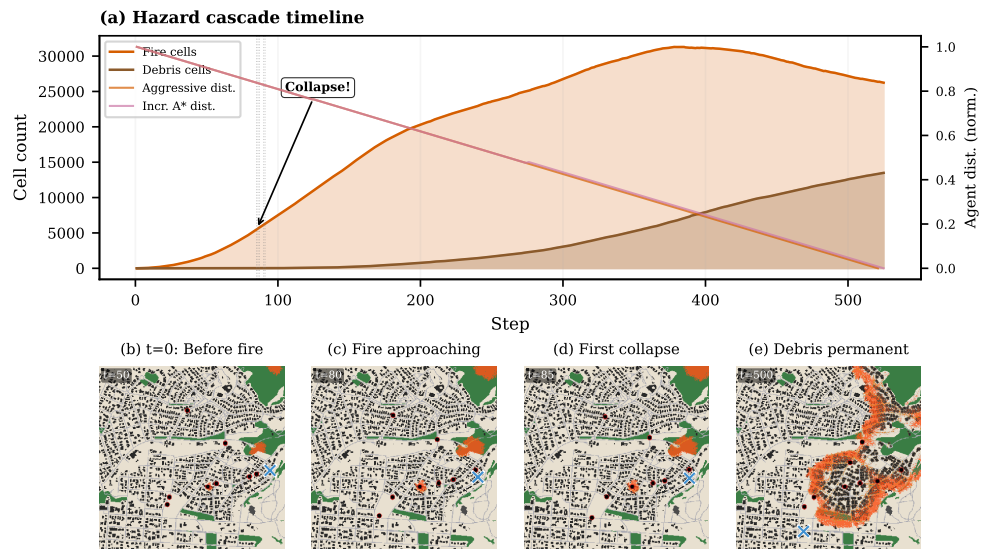


Figure 2. Cascading hazard coupling. (a) Fire peaks then recedes, but debris grows permanently. (b–e) Four snapshots trace the cascade from clean grid to permanent debris field (Penteli, seed 42).

Dynamic obstacles. Moving obstacles patrol the road and maritime network, blocking UAV movement within a Manhattan buffer (radius 5). The interaction engine couples

fire to traffic: fire proximity closes segments, creating compound blockages that neither layer produces alone (Figure 3). 146
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Coupled urban dynamics: emergency vehicles, fire-induced road closures, and corridor blockage



Figure 3. Fire–obstacle coupling (Piraeus, seed 42). (a) Dynamic obstacles operate normally at $t=30$. (b) Fire triggers segment closures by $t=80$. (c) At $t=150$, fire, closures, and diverted obstacles jointly cut the corridor. Black circles denote dynamic obstacles; blue cross denotes the UAV.

2.3. Risk Cost Map 148

The perception pipeline maps raw hazards to planner-specific edge costs in two stages: *risk fusion* and *cost inflation*. 149
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Stage 1: Risk fusion. Five distance-decay layers are fused into a continuous risk map $R : \mathcal{G} \rightarrow [0, 1]$ via element-wise max: 151
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$$R(\mathbf{x}) = \max(R_f, R_s, R_t, R_b, R_{\text{nfz}}) \quad (2)$$

where $R_f = [1 - d_f/r_f]^+$ (fire), $R_s = s/2$ (smoke), $R_t = 0.6[1 - d_t/r_t]^+$ (traffic), $R_b = 0.7[1 - d_b/r_b]^+$ (debris), and $R_{\text{nfz}} \in \{0, 0.5, 1\}$ (no-fly zones), for $\mathbf{x} \in \mathcal{G}$, where d_f, d_t, d_b are Euclidean distance transforms from fire, traffic, and debris with influence radii $r_f=30, r_t=10, r_b=8$ cells; $s \in [0, 1]$ is smoke concentration; $[\cdot]^+ = \max(\cdot, 0)$. The scaling constants (0.6, 0.7) are calibrated heuristics reflecting the relative severity of each hazard layer (insensitive to $\pm 20\%$ perturbation). We use max rather than summation so that a single severe hazard dominates regardless of benign neighbours. 153
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Stage 2: Cost inflation. Each planner inflates its edge cost using a planner-specific risk coefficient $\rho \in \mathbb{R}_{\geq 0}$: 160
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$$w(\mathbf{x}) = 1 + \rho \cdot R(\mathbf{x}) \quad (3)$$

High ρ amplifies perceived danger, causing wider detours; low ρ allows routes through fire fringes. 162
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Figure 4 illustrates that ρ alone—rather than the algorithmic family—determines the effective cost landscape: identical fire state produces five distinct cost surfaces. 164
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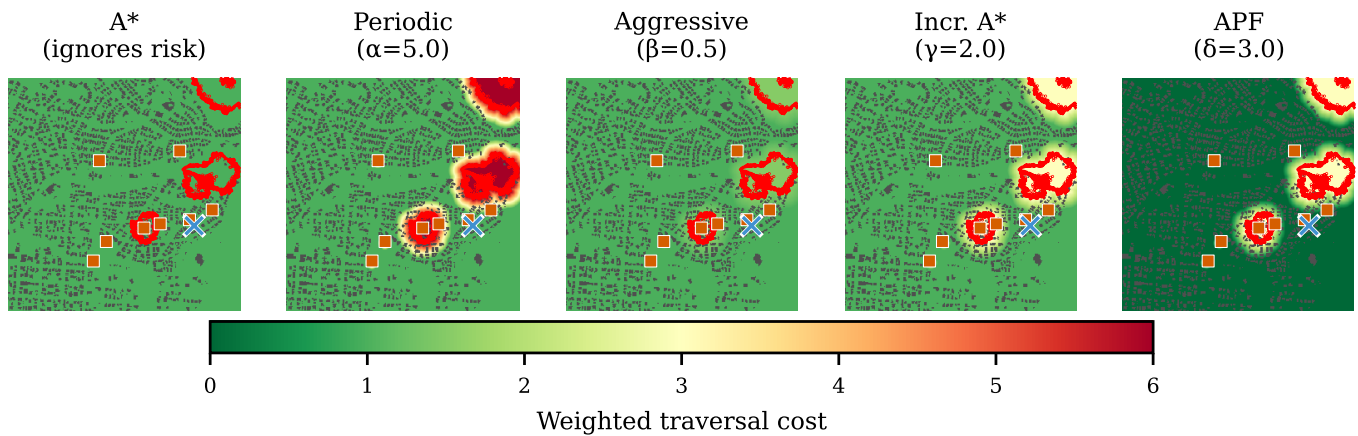
Risk perception at $t=150$: same fire, five different worlds

Figure 4. Planner-specific cost landscapes at $t=150$ for identical fire state. Each panel shows the inflated edge cost $w(\mathbf{x}) = 1 + \rho \cdot R(\mathbf{x})$. A* ($\rho=0$) perceives uniform cost. Periodic ($\alpha=5.0$) inflates fire proximity to near-impassable levels. Aggressive ($\beta=0.5$) applies minimal inflation. The risk coefficient determines the effective cost landscape independently of the planning algorithm.

2.4. Planner Suite

Five planners from three algorithmic families are controlled by a single coefficient ρ (Table 1).

- **A*** [6]: static baseline; plans once, $\rho=0$ (ignores risk).
- **Periodic**: replans every 6 steps, $\alpha=5.0$ (risk-averse).
- **Aggressive**: replans on mask change, $\beta=0.5$ (risk-tolerant).
- **Incremental A*** (D* Lite-inspired [7]): replans on path invalidation, $\gamma=2.0$.
- **APF** [8]: gradient repulsion, $\delta=3.0$.

Table 1. Planner profiles and ranking discrepancy. The Δ column quantifies the divergence between navigation and mission rankings: the highest-SR planner (Incr. A*, Nav.R #1) ranks #4 in mission effectiveness, while the highest-MS planner (Aggressive, Ms.R #1) ranks #3 in navigation. SR = feasible success rate; MS = normalised mission score.

Planner	Family	Risk Coeff.	Replan Trigger	SR%	MS	Nav.R	Ms.R
A*	Static	—	Never	0.0	0.00	5	5
Periodic	Adaptive	$\alpha=5.0$	Every 6 steps	55.4	0.98	2	2
Aggressive	Adaptive	$\beta=0.5$	On obstacle change	50.6	0.99	3	1
Incr. A*	Adaptive	$\gamma=2.0$	On path blocked	62.7	0.54	1	4
APF	Reactive	$\delta=3.0$	Every step (gradient)	33.7	0.57	4	3

2.5. Mission Scoring

Standard benchmarks count successes. FLARE measures the *value* of each success by defining a mission score $M(t) \in [0, 1]$ that is *strictly decreasing* in delivery time $t \in \mathbb{N}$ (Figure 5). Three mission types specialise this framework:

Pharma delivery (Penteli): medication efficacy decays quadratically, $E(t) = \max(0, 1 - (t/T)^2)$, $T=T_{\max}$. The quadratic form reflects pharmaceutical degradation kinetics: insulin and temperature-sensitive medications retain near-full potency during an initial plateau before efficacy degrades sharply as the therapeutic window closes—a profile ill-suited to linear approximation.

Urban rescue (Piraeus): casualty survival decays exponentially with fire-coupled amplification:

$$S_i(t) = \exp(-\lambda_{\text{eff},i} \cdot t), \quad \lambda_{\text{eff}} = \lambda_0(1 + \kappa / \max(d_{\text{fire}}, 1)) \quad (4)$$

where λ_0 is the base decay rate and $\kappa=5.0$ is the fire-coupling constant. Casualties near active fire expire faster, explicitly rewarding planners willing to enter dangerous zones. The fire-coupling term κ/d_{fire} captures a documented triage reality: casualties near an advancing fire front face compounding threats—heat exposure, smoke inhalation, structural instability—that accelerate physiological decline beyond the baseline trauma rate. Consequently, two casualties of identical severity at the same elapsed time may present vastly different rescue values depending solely on fire proximity at the moment of contact.

Fire surveillance (Downtown): data freshness decays linearly, $F(t) = 1 - t/T$. A fire perimeter map becomes stale as the front advances. Linear freshness decay is standard in time-critical intelligence, surveillance, and reconnaissance (ISR) doctrine, where situational awareness degrades at a roughly constant rate as the observed scene evolves.

Key property. All three functions share one characteristic: they are strictly decreasing in t . Every additional step of detour *destroys mission value*. Cross-scenario comparison uses per-scenario normalisation (divide by scenario max, then average).

2.6. Risk–Time–Mission Coupling

The link between risk aversion and mission failure is now explicit. Increasing ρ produces safer paths with higher survival probability but longer travel time t . Because every mission score $M(t)$ is strictly decreasing in t , the additional travel time directly reduces mission value. This creates a fundamental trade-off:

$$\underbrace{\rho \uparrow}_{\text{more risk-averse}} \implies \underbrace{t \uparrow}_{\text{longer detour}} \implies \underbrace{M(t) \downarrow}_{\text{worse mission outcome}} \quad (5)$$

Crucially, the planner receives only a target coordinate and a cost map—it has no access to POI locations, decay functions, or mission-level objectives. The runner orchestrates POI sequencing externally: upon task completion it issues a new target, and upon timeout or abandonment it redirects the planner toward the goal. The ranking inversion is therefore not an artefact of mission-aware planning logic but an emergent consequence of the cost-inflation parameter ρ propagating through path geometry into arrival time and, ultimately, into mission value.

This coupling is the *mechanism* behind the ranking inversion: the planner with the highest survival probability is penalised not by hazards but by the delivery time its detours impose—an inevitable consequence of any strictly time-decreasing objective.

2.7. Scenarios

Each scenario is grounded in a real Greek wildfire event (Table 2).

- **Penteli** (Evia 2021): wind-driven fire in a wildland–urban interface; the mission is insulin delivery before efficacy degrades.
- **Piraeus** (Rhodes 2023): dense port with structural collapse; three casualties with fire-coupled survival.
- **Downtown** (Evros 2023): extreme building density (0.50) with NFZ corridors for manned aircraft.

Each scenario is paired with a mission type that mirrors the dominant operational challenge of its historical precedent. Penteli maps the wildland–urban interface of the 2021 Evia megafire, where isolated settlements required time-critical medical resupply by air af-

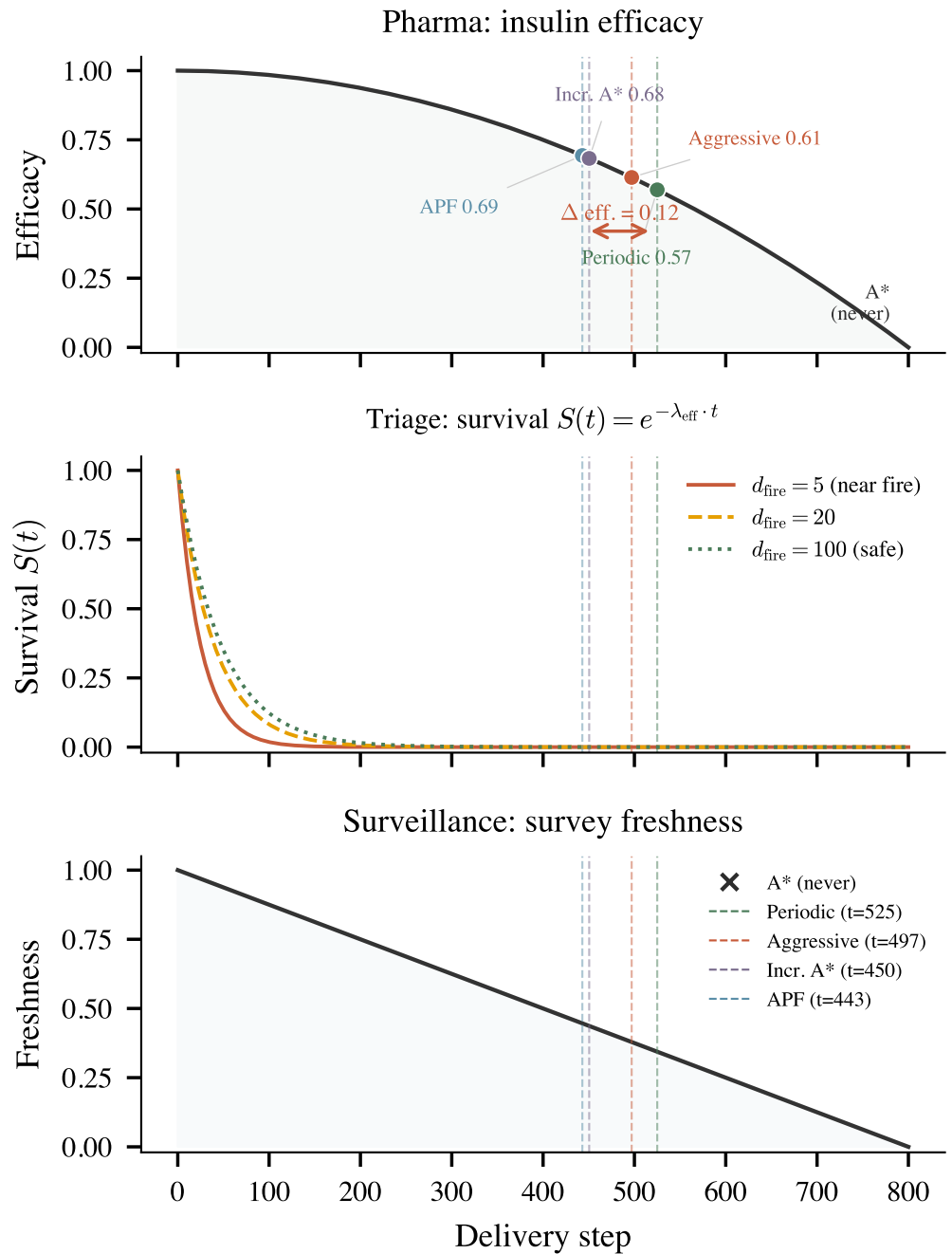


Figure 5. Why speed matters. All three mission scores $M(t)$ decay with delivery time. The shaded gap between Aggressive (low t) and Periodic (high t) quantifies the $M(t)$ cost of a risk-averse detour.

ter road access was severed. Piraeus reproduces the dense port-adjacent topology of the 2023 Rhodes coastal fires, in which structural collapse and segment closures compounded fire hazards, stranding casualties across a fragmented urban grid. Downtown Athens reflects the 2023 Evros megafire—the largest single wildfire in EU history—where overlapping manned-aircraft corridors imposed dynamic no-fly zones on unmanned surveillance operations. These pairings ensure that the decay function, hazard dynamics, and map topology reinforce one another rather than varying independently.

2.8. Reproducibility

FLARE enforces 36 contracts across 15 families, verified by 189 automated tests. Key guarantees: bit-identical replay from a single RNG seed (DC-1), planner-agnostic corridor

Table 2. Scenario overview. Three OSM-based scenarios grounded in real Greek wildfires.

	Penteli	Piraeus	Downtown
Real incident	Evia 2021	Rhodes 2023	Evros 2023
Mission	Insulin delivery	Search & rescue	Fire perimeter survey
Service time	0 (fly-through)	2 steps (hover)	3 steps (hover)
Score decay	Quadratic	Exponential + fire	Linear
Map density	0.18	0.29	0.50
Fire ignitions	2	2	1
Vehicles	8 network + 2 patrol	8 network + 2 patrol	8 network + 2 patrol
NFZ zones	1	1	2
Infeasible seeds	0/30	3/30	4/30
Key challenge	Wind-driven fire sweeps WUI	Collapse + segment closures	Extreme density + NFZ

interdictions (FC-1), frozen hazard state during planning (FD-3). Infeasible episodes are detected at reset and excluded uniformly across all planners.

3. Results

3.1. Experimental Setup

Five planners $\times$ three scenarios $\times$ 30 seeds yield 450 episodes; 7.8% are structurally infeasible and excluded. Statistics follow Agarwal et al. [24]: IQM with 95% bootstrap CIs (10,000 resamples), Wilcoxon signed-rank with Bonferroni correction, Cliff's delta, and Friedman tests.

3.2. Navigation Performance

A* never succeeds: corridor dynamic obstacles block the reference path before fire even arrives—a consequence of ignoring risk entirely. Among adaptive planners, Incremental A* leads (62.7% feasible SR), followed by Periodic (55.4%), Aggressive (50.6%), and APF (33.7%).

Table 3 disaggregates by scenario with mission-specific metrics and verdicts. Two patterns stand out: (i) Piraeus yields the highest success rates (67–70%) because its dense layout provides many alternative routes; (ii) Downtown is hardest, with extreme density constraining detours.

3.3. The Ranking Inversion

Table 1 contains the paper's central result. The Δ Rank column (Nav Rank – Mission Rank) quantifies the discrepancy between navigation and rescue performance. A *ranking inversion* is defined as the condition $\arg \max_{\pi} \text{SR}(\pi) \neq \arg \max_{\pi} M(\pi)$, where SR is feasible success rate and M is normalised mission score. This condition holds across all three scenarios and is confirmed by Friedman test ($p < 0.001$), establishing that navigation success rate does not serve as a reliable proxy for mission effectiveness.

High-SR planner, low mission rank. Incremental A* ranks #1 in SR but only #4 in mission score ($\Delta = -3$). Its efficient, minimal replanning reaches the goal with high probability but delivers medication after significant efficacy loss due to conservative routing. The low replan count (7.5 mean) indicates that the planner does not re-evaluate route quality as the fire state evolves.

Low-SR planner, high mission rank. Aggressive Replan ranks #3 in SR but #1 in mission score ($\Delta = +2$). Its low coefficient ($\beta = 0.5$) admits routes through fire fringes: when the planner survives, medication retains high efficacy ($E(t) > 0.7$) and casualties remain viable ($S(t) > 0$). Low ρ yields shorter paths ($t \downarrow$), directly increasing $M(t)$.

The mechanism follows directly from the coupling in Section 2.6: Aggressive applies $w = 1 + 0.5R$, barely inflating costs near fire, while Periodic applies $w = 1 + 5.0R$ —a $10\times$ stronger aversion signal that produces substantially longer detours. The resulting temporal penalty reduces mission value despite the higher survival probability. Figure 6(a) illustrates this effect: planners on the SR-optimal frontier (right side) fall below the diagonal in mission score, confirming the ranking discrepancy.

Table 3. Per-scenario results with mission-specific metrics. $\star$ denotes the highest mission score per scenario; **bold** denotes the highest success rate. The two markers fall on different planners in each scenario, consistent with the ranking inversion.

Planner	SR%	Delivery Step	Mission Metric	Verdict
Penteli — Insulin delivery to fire-isolated village (Evia 2021)				
A*	0	timeout	0%	Collision with dynamic obstacle
Periodic	40	~514	59%	Extended detour, efficacy loss
Aggressive	33	~429	71%	Short route via smoke corridor
Incr. A*	53	~408	$\star$ 74%	Highest SR, timely delivery
APF	23	~412	73%	Local-minimum delays
Piraeus — Urban rescue in port district (Rhodes 2023)				
A*	0	timeout	0.00	Blocked by fire and collapse
Periodic	70	~477	2.78	High SR, delayed casualty contact
Aggressive	67	~486	$\star$ 2.98	Early casualty contact
Incr. A*	67	~480	1.62	Moderate navigation and rescue
APF	48	~392	1.82	Local-minimum induced delays
Downtown — Fire perimeter survey (Evros 2023)				
A*	0	timeout	0%	No feasible static path
Periodic	58	~599	25%	Conservative route, data staleness
Aggressive	54	~563	30%	Shorter route, moderate risk
Incr. A*	69	~447	$\star$ 44%	Fastest survey, highest freshness
APF	31	~588	26%	Gradient drift, slow coverage

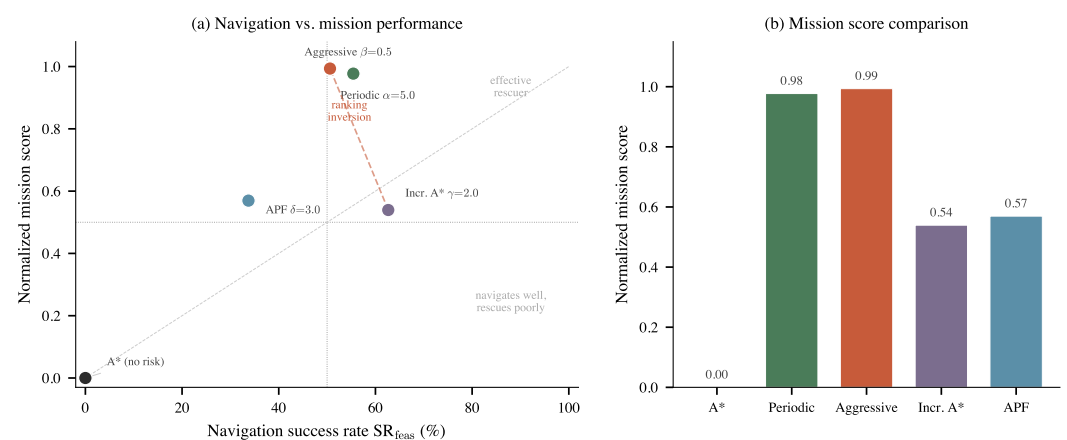


Figure 6. Ranking inversion between navigation and mission metrics. (a) Success rate vs. normalised mission score; the dashed line connecting Incr. A* to Aggressive crosses the diagonal, indicating divergent orderings. (b) Per-planner mission-score decomposition confirms that Aggressive achieves the highest aggregate score despite a lower success rate.

3.4. Qualitative Evidence

Figure 7 provides spatial confirmation of the ranking inversion. Under identical fire conditions, three planners produce qualitatively different trajectories. A* terminates upon collision with a dynamic obstacle at $t=78$. Aggressive and Periodic both complete the mission in comparable step counts (522 vs. 524) but via substantially different paths: Aggressive traverses the smoke corridor while Periodic circumnavigates the fire perimeter along a path approximately twice as long. The risk coefficient, rather than the algorithmic family, determines route geometry.

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Figure 7. Trajectory comparison under identical fire conditions (Penteli, seed 42). A* (left) terminates upon collision with a dynamic obstacle. Aggressive (centre) traverses the smoke corridor. Periodic (right) completes a longer detour, arriving with reduced medication efficacy. Black circles: dynamic obstacles; blue cross: UAV position; yellow cross: target.

3.5. Statistical Validation

Wilcoxon tests (Bonferroni-corrected) confirm A* differs from every adaptive planner ($p < 0.001$). Critically, *no significant difference* exists between Periodic and Aggressive in SR ($p = 1.0$): navigation success alone cannot separate them. The discriminator is mission score: a Friedman test yields $\chi^2(4) = 190.8$, $p < 0.001$, establishing the ranking inversion as statistically robust (Table 4).

Table 4. Statistical validation. Friedman confirms ranking differences; bootstrap CIs quantify uncertainty.

Test	Statistic	p	Result
Friedman (5 planners)	$\chi^2=190.8$	<0.001	Planners differ
A* vs. Adaptive (Wilcoxon)	—	<0.001	All significant
A* vs. Incr. A* (Cliff's δ)	0.58	—	Large effect
Incr. A* vs. APF (Cliff's δ)	0.27	—	Small effect
<i>IQM success rate, 95% bootstrap CI (10,000 resamples):</i>			
Incr. A*	62.7%	[50.9, 74.5]	
Periodic	55.4%	[43.1, 67.8]	
Aggressive	50.6%	[38.6, 62.6]	
APF	33.7%	[22.6, 44.9]	

3.6. Hazard Ablation

Isolating dynamics layers reveals their contributions. Fire is the dominant differentiator: it reduces A* to 0% while adaptive planners retain 39–64%. Traffic alone also eliminates A* but has near-zero marginal impact on Periodic and Aggressive—their replanning fully mitigates obstacle blockages. Combined dynamics reduce all planners by 10–20 pp relative to fire alone, confirming non-additive layer interaction. Road closures grow proportionally with fire (via the interaction engine); debris accumulates after the collapse delay ($\tau_c=80$). Aggressive Replan's timely detours maintain forward progress; Periodic's risk-averse strategy produces extended stalls—a direct consequence of the ρ - t - M coupling.

4. Discussion

4.1. Shapley Attribution

Exact Shapley values [32] over coalitions {fire, traffic, collapse} confirm fire as the dominant hazard for all planners. Traffic degrades A* ($\phi = -0.50$) and APF ($\phi = -0.25$) but

has zero marginal impact on Periodic and Aggressive—their replanning absorbs obstacle blockages entirely. Collapse contributes near-zero marginal effect, subsumed by the fire that triggers it.

In fire-dominated scenarios, the risk coefficient ρ has greater influence on mission outcome than replanning frequency; in traffic-heavy environments, all adaptive planners perform comparably.

4.2. Risk Coefficient as a Mission-Selection Parameter

The ranking inversion establishes that no universally optimal planner exists across mission types. Instead, ρ serves as a principled mission-selection parameter:

- **Time-critical delivery** (rescue, medication): $\rho \leq 0.5$. Minimise t to maximise $M(t)$; accept higher collision probability.
- **Survival-critical operations** (surveillance): $\rho \geq 5.0$. Delayed data retains partial value; collision yields $M=0$.

This finding reframes planner selection: rather than identifying a single superior algorithm, the appropriate risk coefficient is determined by the mission-specific relationship between delivery time and objective value—a relationship that FLARE makes empirically quantifiable.

4.3. Limitations

Several simplifications constrain the scope of the present study. FLARE operates on 2D grids without altitude modelling or aerodynamic constraints. The planner suite comprises classical algorithms; the CA fire model provides an approximation of real wildfire dynamics. Three Greek urban scenarios offer focused but not exhaustive geographic coverage. Risk coefficients are fixed per planner and not adapted online. The ranking inversion was not predictable *a priori*, as no prior benchmark incorporates time-coupled mission scoring, and the magnitude ($|\Delta\text{Rank}|=3$) required empirical measurement. Importantly, ρ is not a tunable hyperparameter but a structural property of each planner's cost function, and the inversion persists across all three geographically distinct scenarios. These simplifications are deliberate: the contribution is the *evaluation paradigm* and the *ranking-inversion finding*, not the planners themselves.

5. Conclusions

This paper presented FLARE, a deterministic benchmark designed to evaluate UAV planner performance under time-coupled mission objectives. Across 450 episodes on three Greek wildfire scenarios, the experimental results establish a ranking inversion: the planner with the highest survival probability ranks last in mission score $M(t)$, because its detours increase delivery time t beyond the point at which the objective retains value. Conversely, a risk-tolerant planner achieves the highest mission score despite ranking third in navigation success rate (Friedman $p < 0.001$).

The ranking inversion demonstrates that navigation success rate constitutes an insufficient evaluation criterion for time-critical operations, as it can produce orderings that are inversely correlated with mission effectiveness. The principal implication is methodological: planner evaluation should incorporate time-dependent mission value functions rather than relying solely on binary completion metrics. The risk coefficient ρ provides a principled parameter for mission-specific planner selection. The benchmark framework and all experimental data are publicly available.

Code and data: <https://github.com/zervakisai/uavplan>.

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Data Availability Statement: The benchmark code, scenario configurations, and full experimental results are publicly available at <https://github.com/zervakisai/uavplan>. Map data are derived from OpenStreetMap [33] under the Open Database License (ODbL).

Acknowledgments: Map data © OpenStreetMap contributors (ODbL). Scenarios inspired by the 2021 Evia wildfire [34,35], 2023 Rhodes fires [36], and 2023 Evros megafire [37,38]. Data from EFFIS [34,38] and Copernicus EMS [35–37].

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Abbreviations

The following abbreviations are used in this manuscript:

UAV	Unmanned Aerial Vehicle
FLARE	Fire-Landscape Adaptive Risk Evaluation
SR	Success Rate
IQM	Interquartile Mean
CI	Confidence Interval
CA	Cellular Automaton
APF	Artificial Potential Field
OSM	OpenStreetMap
NFZ	No-Fly Zone
RNG	Random Number Generator
WUI	Wildland–Urban Interface

References

1. Erdelj, M.; Natalizio, E.; Chowdhury, K.R.; Akyildiz, I.F. Help from the Sky: Leveraging UAVs for Disaster Management. *IEEE Pervasive Computing* **2017**, *16*, 24–32. <https://doi.org/10.1109/MPRV.2017.11>.
2. Yücesoy, G.; et al. The Role of drones in disaster response: A literature review of operations research applications. *International Transactions in Operational Research* **2025**, *32*, 545–589. <https://doi.org/10.1111/itor.13484>.
3. Rahim, F.; Qasim, N.H. A systematic literature review of drones in emergency medicine: Practical applications, legal challenges, and future directions. *Drone Systems and Applications* **2025**, *13*, 1–13. <https://doi.org/10.1139/dsa-2024-0072>.
4. Li, J.; Pei, Y.; Zhao, S.; Xiao, R.; Sang, X.; Zhang, C. A comprehensive survey of research towards AI-enabled unmanned aerial systems in pre-, active-, and post-wildfire management. *Information Fusion* **2024**, *108*, 102369. <https://doi.org/10.1016/j.inffus.2024.102369>.
5. Dradoun, A.; Khelassi, A.; Lachekhab, F. Intelligent Path Planning Algorithms for UAVs: Classification, Complexity Analysis, Hybrid Ablation Insights, and Future Directions. *Advances in Mechanical Engineering* **2025**, *17*. <https://doi.org/10.1177/16878132251355020>.
6. Hart, P.E.; Nilsson, N.J.; Raphael, B. A Formal Basis for the Heuristic Determination of Minimum Cost Paths. *IEEE Transactions on Systems Science and Cybernetics* **1968**, *4*, 100–107. <https://doi.org/10.1109/TSSC.1968.300136>.
7. Koenig, S.; Likhachev, M. D* Lite. In Proceedings of the Proceedings of the 18th AAAI Conference on Artificial Intelligence, 2002, pp. 476–483.
8. Khatib, O. Real-Time Obstacle Avoidance for Manipulators and Mobile Robots. *The International Journal of Robotics Research* **1986**, *5*, 90–98. <https://doi.org/10.1177/027836498600500106>.
9. Meng, W.; Zhang, X.; Zhou, L.; Guo, H.; Hu, X. Advances in UAV Path Planning: A Comprehensive Review of Methods, Challenges, and Future Directions. *Drones* **2025**, *9*, 376. <https://doi.org/10.3390/drones9050376>.
10. Aggarwal, S.; Kumar, N. Path Planning Techniques for Unmanned Aerial Vehicles: A Review, Solutions, and Challenges. *Computer Communications* **2020**, *149*, 270–299. <https://doi.org/10.1016/j.comcom.2019.10.014>.
11. Patra, S.; Basak, S.; Patel, H. A Survey on Reinforcement Learning Methods for UAV Systems. *ACM Computing Surveys* **2025**, *57*, 1–42. <https://doi.org/10.1145/3769426>.
12. Bouzid, Y.; Bestaoui Sebbane, Y.; Mohammedi, A. UAV Path Planning Trends from 2000 to 2024: A Bibliometric Analysis and Visualization. *Drones* **2025**, *9*, 128. <https://doi.org/10.3390/drones9020128>.
13. Sánchez-Ibáñez, J.R.; Pérez-del Pulgar, C.J.; García-Cerezo, A. Comprehensive Review of Path Planning Techniques for Unmanned Aerial Vehicles (UAVs). *ACM Computing Surveys* **2025**, *57*, 1–41. <https://doi.org/10.1145/3737280>.
14. Zhou, W.; et al. A risk-based unmanned aerial vehicle path planning scheme for complex air-ground environments. *Risk Analysis* **2025**, *45*, 321–342. <https://doi.org/10.1111/risa.17685>.
15. Melissaris, C.; et al. Risk-aware UAV trajectory optimization using open urban GIS data and target level of safety constraints. *Drones* **2025**, *9*, 666. <https://doi.org/10.3390/drones9100666>.

16. Karaaç, C.; Ylmaz, N.; Hançerlioullar, A. Risk-Aware Enabled Path Planning for Drones Flight in Unknown Environment. *Journal of Intelligent & Robotic Systems* **2025**, *111*, 56. <https://doi.org/10.1007/s10846-025-02243-2>. 397
17. Chen, X.; Zhang, F.; Liu, Y.; Zhang, L. Probabilistic Path Planning for UAVs in Forest Fire Monitoring: Enhancing Patrol Efficiency through Risk Assessment. *Fire* **2024**, *7*, 254. <https://doi.org/10.3390/fire7070254>. 398
18. Alexandridis, A.; Vakalis, D.; Siettos, C.I.; Bafas, G.V. A Cellular Automata Model for Forest Fire Spread Prediction: The Case of the Wildfire that Swept Through Spetses Island in 1990. *Applied Mathematics and Computation* **2008**, *204*, 191–201. <https://doi.org/10.1016/j.amc.2008.06.046>. 399
19. Giannino, F.; et al. Statistical Dynamics of Wildfire Burned Area from Cellular-Automata Simulators. *Scientific Reports* **2025**, *15*, 2025. <https://doi.org/10.1038/s41598-025-30851-3>. 400
20. Sahila, A.; Fiorese, A.; Perello, N.; Trucchia, A.; Pagnini, G. Patterns of burned area by a cellular-automata fire simulator: The role of microscale wind field. *Communications in Nonlinear Science and Numerical Simulation* **2025**, *151*, 109026. <https://doi.org/10.1016/j.cnsns.2025.109026>. 401
21. Wang, Z.; Zhao, M.; Chen, G.; Li, J. Multi-Target Firefighting Task Planning Strategy for Multiple UAVs Under Dynamic Forest Fire Environment. *Fire* **2025**, *8*, 61. <https://doi.org/10.3390/fire8020061>. 402
22. Liu, Y.; Wang, Z.; Chen, X.; Zhang, W. Multiple UAV Swarms Collaborative Firefighting Strategy Considering Forest Fire Spread and Resource Constraints. *Drones* **2025**, *9*, 17. <https://doi.org/10.3390/drones9010017>. 403
23. Henderson, P.; Islam, R.; Bachman, P.; Pineau, J.; Precup, D.; Meger, D. Deep Reinforcement Learning That Matters. In Proceedings of the Proceedings of the 32nd AAAI Conference on Artificial Intelligence, 2018. 404
24. Agarwal, R.; Schwarzer, M.; Castro, P.S.; Courville, A.; Bellemare, M.G. Deep Reinforcement Learning at the Edge of the Statistical Precipice. In Proceedings of the Advances in Neural Information Processing Systems, 2021, Vol. 34. 405
25. Demšar, J. Statistical Comparisons of Classifiers over Multiple Data Sets. *Journal of Machine Learning Research* **2006**, *7*, 1–30. 406
26. Dolan, E.D.; Moré, J.J. Benchmarking Optimization Software with Performance Profiles. *Mathematical Programming* **2002**, *91*, 201–213. <https://doi.org/10.1007/s101070100263>. 407
27. Bonsignorio, F.; del Pobil, A.P.; Zereik, E. Towards reproducible robotics research. *Nature Machine Intelligence* **2025**, *7*, 15–22. <https://doi.org/10.1038/s42256-025-01114-7>. 408
28. Ferrara, C.; et al. Statistical dynamics of wildfire burned area from cellular-automata simulators. *Scientific Reports* **2025**. <https://doi.org/10.1038/s41598-025-30851-3>. 409
29. Zhao, Y.; Zheng, Z.; Liu, Y. Survey on Computational-Intelligence-Based UAV Path Planning. *Knowledge-Based Systems* **2018**, *158*, 54–64. <https://doi.org/10.1016/j.knosys.2018.05.033>. 410
30. Boeing, G. OSMnx: New Methods for Acquiring, Constructing, Analyzing, and Visualizing Complex Street Networks. *Computers, Environment and Urban Systems* **2017**, *65*, 126–139. <https://doi.org/10.1016/j.compenvurbsys.2017.05.004>. 411
31. Towers, M.; Kwiatkowski, A.; Terry, J.K.; Balis, J.U.; De Cola, G.; Deleu, T.; Goulão, M.; et al. Gymnasium: A Standard Interface for Reinforcement Learning Environments. In Proceedings of the Advances in Neural Information Processing Systems, 2025, Vol. 38. Spotlight. 412
32. Shapley, L.S. A Value for n-Person Games. In *Contributions to the Theory of Games II*; Kuhn, H.; Tucker, A., Eds.; Princeton University Press, 1953; pp. 307–317. 413
33. OpenStreetMap contributors. OpenStreetMap, 2024. Data available under the Open Database License (ODbL). 414
34. San-Miguel-Ayanz, J.; Durrant, T.; Boca, R.; Maianti, P.; Libertá, G.; Artés-Vivancos, T.; Oom, D.; Branco, A.; Rigo, D. Forest Fires in Europe, Middle East and North Africa 2021. Technical report, European Commission, Joint Research Centre, 2022. EUR 31094 EN, <https://doi.org/10.2760/34094>. 415
35. Copernicus Emergency Management Service. Copernicus EMS Rapid Mapping: Wildfires in Evia and Attica, Greece, August 2021, 2021. Over 50,000 ha burned on Evia island; largest wildfire recorded in EU at the time. 416
36. Copernicus Emergency Management Service. Copernicus EMS Rapid Mapping Activation EMSR675: Wildfire in Rhodes, Greece, 2023. 18 monitoring maps delivered 20–30 July 2023; 17,629 ha burned on Rhodes island. 417
37. Copernicus Emergency Management Service. Copernicus EMS Risk and Recovery Activation EMSN166: Wildfire in Evros, Greece, 2023. Evros fire started 19 August 2023 near Aristino village, 96,000+ ha burned — largest single wildfire in EU history. 418
38. San-Miguel-Ayanz, J.; et al. Forest Fires in Europe, Middle East and North Africa 2023. Technical report, European Commission, Joint Research Centre, 2024. Covers Evros megafire (96,000 ha) and Rhodes fires; EUR 32234 EN, <https://doi.org/10.2760/108466>. 419